## Summary

- Vitamins and Minerals are micronutrients which are present in small quantities. They help to protect body from diseases.




- They are broadly divided into two categories, on the basis of solubility in fats or water. Fat soluble vitamins - A, D, E and K. Fat soluble vitamins are stored in the liver and fatty tissues. These are not readily excreted from the body. Water soluble vitamins - B ($B_1$, $B_2$, $B_3$, $B_5$, $B_6$, $B_{12}$) and C. Water soluble vitamins are stored in limited amounts and are readily excreted from the body through urine.
- Vitamin A occurs only in foods of animal origin. Vitamin A activity is possessed by carotenoids found in plants. Hence carotenoids are called provitamin A.
- Vitamin D is otherwise known as 'sunshine vitamin' as it can be synthesized from sunlight by our body.
- Vitamin E protects cell membranes. They are called as tocopherol. Vitamin K is called coagulation vitamin. It is essential for the production of a type of protein called prothrombin and other factors involved in the blood-clotting mechanism.
- The B vitamins have important metabolic roles as coenzyme partners with cell enzymes that control energy metabolism.
- Vitamin C is also known as ascorbic acid. It is an antioxidant and water soluble vitamin. It is destroyed in light, heat and when exposed to air and metals.
- The body contains about 24 minerals, all of which must be provided by the diet. These are required by the body in very minute amounts and are often referred to as trace elements. The important ones are iron, iodine, calcium, zinc and sodium.
- Water is vital for human existence. Water is the largest component of the human body. Normal intake of water ranges between 8 – 10 glasses per day.
- Dehydration results in extreme deficiency of water and fluids. Oral rehydration therapy is a method of treating dehydration by making the patient drink a solution which can be prepared by dissolving salt and sugar in boiled and cooled water.

## Glossary

| Terms | Meaning |
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| Coenzymes | A compound that is essential for the functioning of an enzyme. |
| Enzyme | A substance produced by a living organism that assists in chemical processes. |
| Precursor | A forerunner |
| Nyctalopia | A condition characterized by an abnormal inability to see in dim light or at night, typically caused by vitamin A deficiency. |
| Antioxidant | A substance that counteracts oxidation. |



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